Characterization of Galaxies via Optical and Infrared Surveys

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Abstract

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1 Introduction

With ongoing and upcoming surveys providing ever larger amounts of data, the task of object classification is becoming increasingly relevant. In this work, we will summarize and compare select methods for characterizing narrow emission line galaxies and the extraction of their physical properties. For all steps in the following sections, we use data from the latest SDSS data release and match those to spectra obtained by WISE for the same objects (Wright et al. 2010, Kollmeier et al. 2026). Information to reproduce these results are included in the appendix.

2 Optical & Infrared Classification

Generally, the broad labeling of galaxies as Active Galactic Nuclei (AGN) or Star Formation (SF) dominated types relies on reconstructing the excitation and ionization mechanisms from emission lines, and determining whether these match the thermal continuum of stellar populations or the strongly nonthermal radiation produced in AGN powered by Super Massive Black Holes (SMBH) in their centers. Approaches analyzing the continuum emission provide complementary insights, and by probing different wavelengths we can peer through obscuring dust and gas to unveil the true underlying physics. Now, we will explore some famous examples.

2.1 BPT Diagram

By plotting the ratio of emission line fluxes for [OIII] at 5007 Å to Hβ at 4863 Å versus [NII] at 6583 Å to Hα at 6565 Å in their work, Baldwin, Phillips, and Terlevich draw a diagram that sorts galaxies into a distinctive winged shape (Baldwin et al. 1981). Due to lacking the energies to ionize heavier atoms, SF types tend towards the lower left of the image, while powerful AGN fall into the upper right tip. By choosing emission lines close to one another, this framework automatically cancels out reddening differences as well as calibration errors. Depicted in Figure 1 is our entire dataset as a scatter plot that has been shaded based on the density of nearby points. This convention is adopted for all such diagrams in this work for better interpretability of the data. Together with this, standard theoretical (Kewley et al. 2001) and empirical (Kauffmann et al. 2003) curves to discriminate between AGN and SF classes are shown.

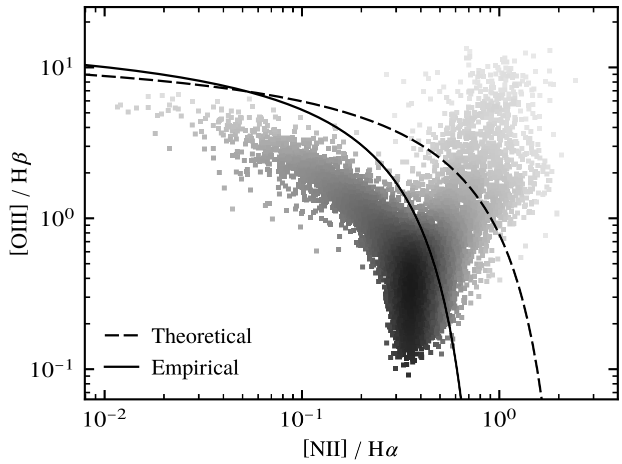

Figure 1: Standard BPT diagram and class separation curves (Kewley et al. 2001, Kauffmann et al. 2003).

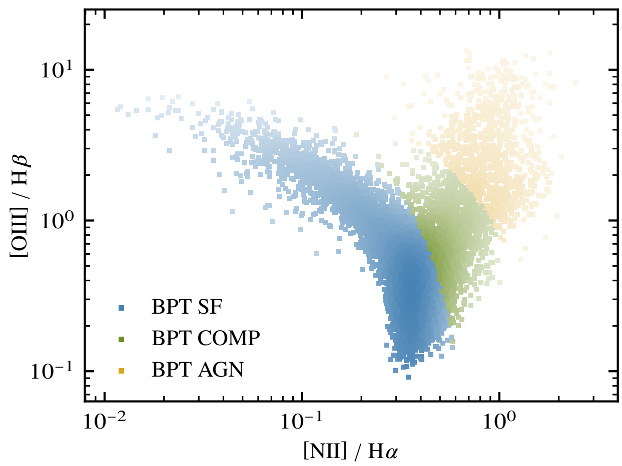

Figure 2: Colored BPT diagram with derived class labels.

We label points between the lines as COMP for composite types and display the classes in Figure 2 encoded by color. Additionally, we have access to the already processed SDSS labels which we can similarly plot in the BPT diagram. This is done in Figure 3 for the AGN and SF as well as starburst SB subclasses. Comparing to the previous plot gives a first visual impression of their cross method agreement, which for the AGN wing seems acceptable, but will be evaluated more rigorously in the coming paragraphs.

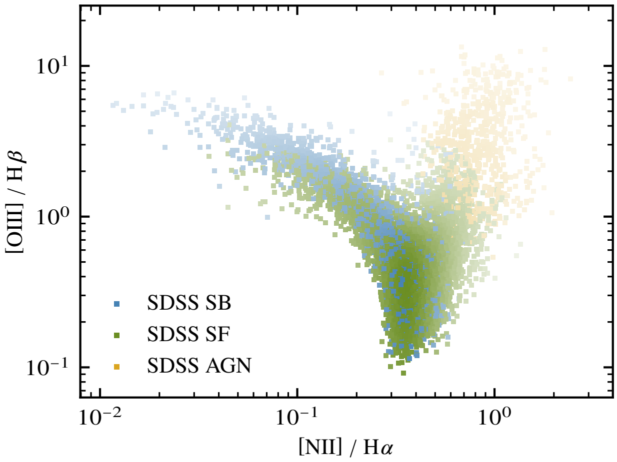

Figure 3: Colored BPT diagram based on SDSS labels.

2.2 WHAN Diagram

Next up is the WHAN diagram, which plots the equivalent width $W_{\mathrm{H}\alpha}$ of the Hα line against the ratio of [NII] to Hα fluxes (Cid Fernandes et al. 2011). This width quantifies line strengths in terms of the underlying continuum and is thereby independent of measurement effects. The concept behind a classification using the measure is that based on the prominence of lines above the background radiation, we can estimate the relative photon budget of the exciting or ionizing radiation field. In true AGN and SF galaxies, this value is high, whereas retired galaxies with old stellar populations have less pronounced emission lines. This is a distinction which the BPT classifier fails to take into account and can therefore improve our label confidence.

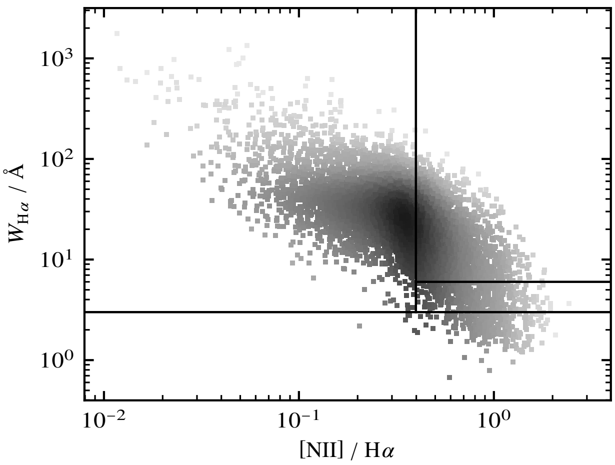

Figure 4: Raw data in WHAN diagram with separators.

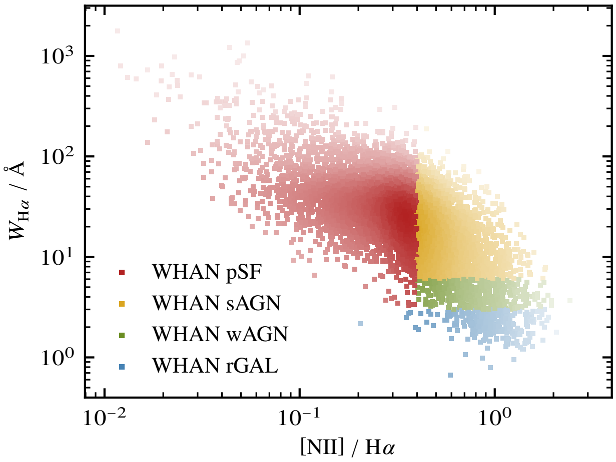

Figure 5: Colored WHAN diagram with derived labels.

With the raw scattered data, Figure 4 includes discrimination lines. According to these, any galaxy with $W_{\mathrm{H}\alpha} < 3.0$ Å is classified as rGAL or retired. Points above this value with $\log_{10}([\mathrm{NII}]\ /\ \mathrm{H}\alpha) > -0.4$ are labeled AGN or else pSF for pure star formation. Within the active class, we also include the distinction for strong sAGN if $W_{\mathrm{H}\alpha} > 6.0$ Å and weak wAGN otherwise. There is further a category defined via $W_{\mathrm{H}\alpha} < 0.5$ Å and $W_{[\mathrm{NII}]} < 0.5$ Å for passive, actually lineless galaxies, but based on our selection, none fall into this class. Figure 5 shows the resulting categories, while Figure 6 presents the more restrictive SDSS classes, which show a smaller AGN region than what WHAN includes.

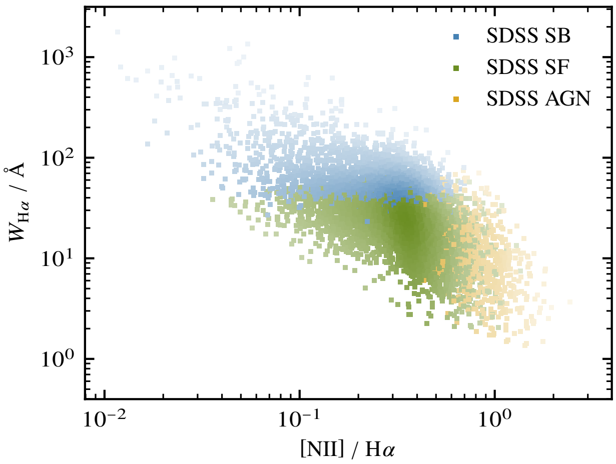

Figure 6: Colored WHAN diagram based on SDSS labels.

2.3 Category Mixing

In addition to the predefined SDSS label, we can check the mixing between BPT and WHAN classes. For this, Figure 7 displays WHAN colors in the BPT diagram, and Figure 8 the opposite. Here we can see how BPT conditions do not acccount for weak equivalent widths, making WHAN a more nuanced model in this regard. On the other hand, WHAN labels many objects AGN that sit far down in the BPT scale, indicating that information on the ionization potential of the source gets lost.

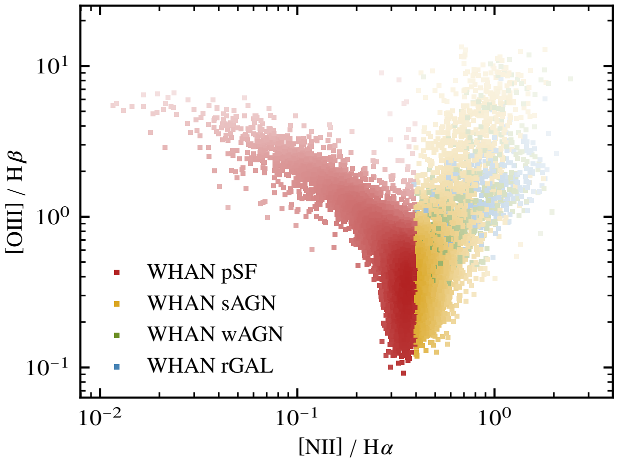

Figure 7: Colored BPT diagram based on WHAN labels.

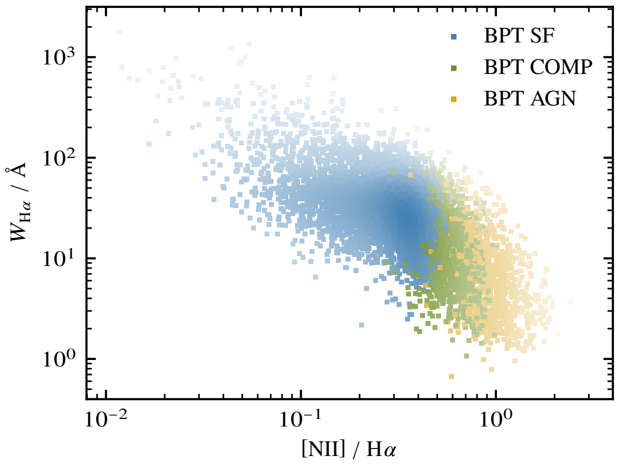

Figure 8: Colored WHAN diagram based on BPT labels.

2.4 WISE Colors

For sources shrouded by dust, optical classifier such as BPT and WHAN can fail. Instead, infrared data from missions such as WISE with filters labeled W1 at 3.4 µm and W2 at 4.6 µm as well as W3 at 12 µm center wavelengths can be used. In these scenarios, galaxies dominated by their stars appear in the more energetic bands corresponding to stellar atmospheres, while an AGN torus absorbs shorter wavelengths, leading to smaller frequencies characterizing their spectrum. In the typical WISE colors plot, we draw the differences of the respective magnitudes, equivalent to the logarithm of the flux ratio and therefore independent of true luminosity. In this diagram, we can define a region where most AGN fall, referred to as the wedge from here on (Mateos et al. 2012). Our data is visualized and classified according to this wedge in Figure 9 with schematic classes from (Wright et al. 2010) taken as a backdrop. Notably, there are only few objects being labeled as AGN by this method. For the actual statistics and a more robust comparison of the classifiers, the following subsection calculates the pairwise binary cross correlation and gives the raw numbers for our matched dataset.

2.5 Evaluating Diagnostics

Listed in Table 1 are the numbers for our processed dataset. To better compare the performance between the different classifiers, we compute the so called Matthews Correlation Coefficient (MCC) for each pair. This metric measures how the matching quality compares to a pure chance baseline. A value of 1 means perfect agreement, while 0 implies no advantage over randomly assigning an answer, and negative values signify worse than random anticorrelation between methods. It also corrects for unbalanced population sizes, as is the case for our sample in which AGN are far fewer than other types of galaxies. Figures 10 and 11 show our results for two cases of BPT classifier, one strict which restricts positives to AGN objects only, and one relaxed that also includes COMP labels.

Table 1: List of overall and relative positive AGN labels by method. In total, 10 000 unique SDSS objects were used, of which 9799 had matches in the WISE database with a 3″ tolerance.

Method	Total	Fraction
SDSS	585	5.9 %
BPT (strict)	920	9.2 %
BPT (relaxed)	2491	24.9 %
WHAN	3464	34.6 %
WISE	88	0.9 %
SDSS & WISE	37	0.4 %
BPT (strict) & WISE	47	0.5 %
BPT (relaxed) & WISE	64	0.6 %
WHAN & WISE	58	0.6 %

We recognize that generally, optical methods agree with amongst each other to a higher degree than with infrared based results. On the other hand, agreement between WISE labels is barely better than random, especially for the WHAN results. For the strict BPT case, it matches best with SDSS classes, while the relaxed version shows better agreement with WHAN labels.

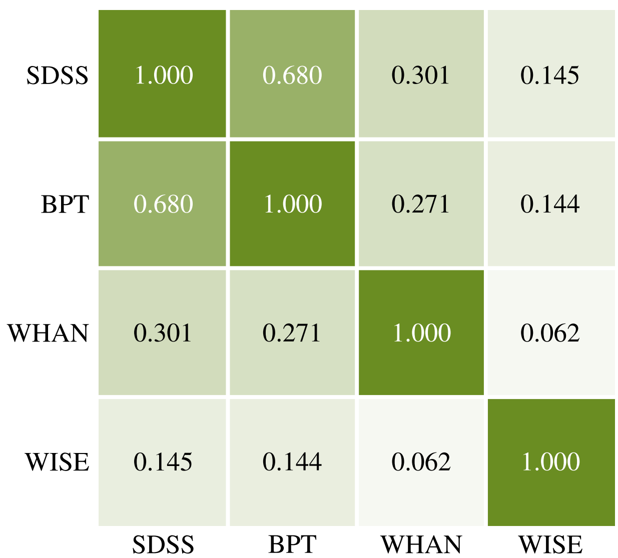

Figure 10: Matrix of MCC results with strict BPT method.

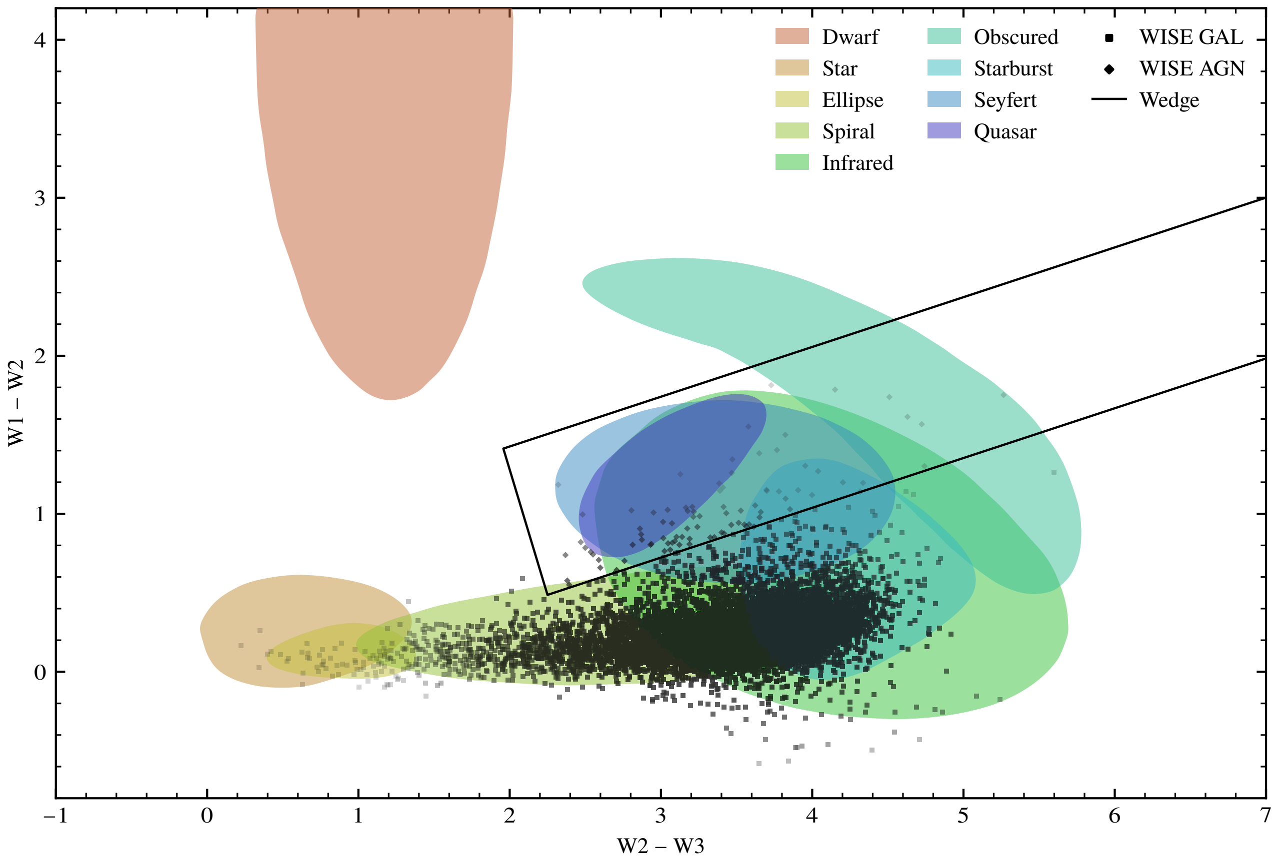

Figure 9: Shaded scatter plot of data in WISE color diagram. Included is the wedge discriminator (Mateos et al. 2012) as well as the background adapted from (Wright et al. 2010) with shortened object type labels.

Based on this, we will proceed in the BPT framework for the following section and deliver SDSS and WHAN as well as WISE results for comparision in the appendix. There are more powerful statistical diagnostics that enable a detailed view of method performance, but these are omitted in this work due to its limited scope. Furthermore, sophisticated classifiers are likely able to achieve physically more sound results, though this comes at the cost of higher complexity.

	SDSS	BPT	WHAN	WISE
SDSS	1.000	0.433	0.301	0.145
BPT	0.433	1.000	0.631	0.104
WHAN	0.301	0.631	1.000	0.062
WISE	0.145	0.104	0.062	1.000

Figure 11: Matrix of MCC results with relaxed BPT method.

3 Physical Properties

Other than classifying objects, we can also extract estimates of their physical properties from spectral data. The following subsections will outline this process for electron density and gas velocity, as well as derive the central Black Hole (BH) mass for our dataset. Graphics in this section use hexagonal bins instead of scatter plots to retain the distribution shape while making possible trends in the data more visible.

3.1 Gas Density

First, we use the ratio of the two [NII] lines at 6716 Å and 6731 Å in our data. Since these both originate from a closely spaced doublet, their relative occurence is independent of gas temperature. Instead, at low densities the ratio is fixed by their degeneracies as statistical weights, whereas at high densities, collisional deexcitation becomes relevant, which does not emit photons. Due to the lower wavelength state having a longer lifetime, it gets affected more strongly by this mechanism, dropping the ratio and allowing us to reconstruct the local electron density as a proxy for the gas (Sanders et al. 2016, Joh et al. 2021). The resulting values are depicted in Figure 12 and show a soft but clearly recognizable trend towards higher densities in the AGN classified region. This makes sense if we imagine that the harsh radiation field as well as shocks originating from the central SMBH compress the surrounding Narrow Line Region (NLR) gas from which our lines are emitting.

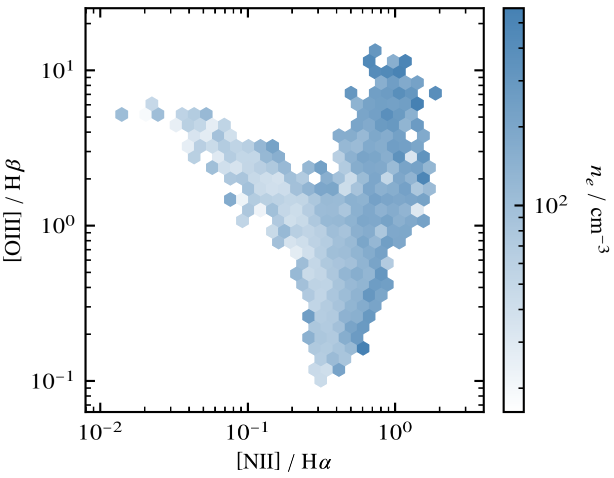

Figure 12: Binned BPT data showing electron density. Refer to Figures 18 and 19 for WHAN and WISE plots.

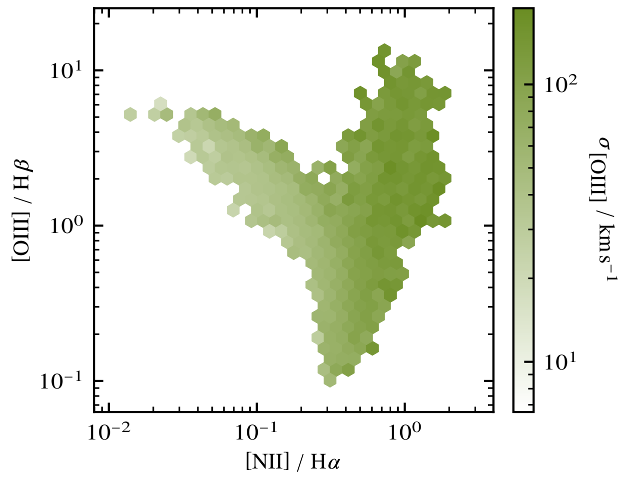

Figure 13: Binned BPT data showing gas velocity. Refer to Figures 20 and 21 for WHAN and WISE plots.

3.2 Gas Velocity

Finding the gas kinematics is comparatively easy. We simply take the [OIII] line dispersion $\sigma_{[OIII]}$ which we assume is caused by Doppler broadening from the random movement of individual gas clouds as the local velocity of turbulence and kinetic energy. We expect this value to be lower for SF type galaxies, in which the gas motions are relatively orderly, following mostly the gravitational rotation of the disk with some stellar feedback present. In AGN on the other hand, the gas is strongly affected by the presence of a deep gravity well and energized through outflows, radio jets, or shocks coming from the central engine. Looking at our data, this is broadly what we observe. Figure 13 shows a clear increase of velocity width towards the upper right AGN wing, while lower ionization states in the left SF tip tend to have the smallest values. Here, it would be interesting to evaluate whether the increased velocities towards the AGN branch are purely a result of the correlation with a more massive central BH and the faster rotation required to match the higher mass, or if AGN feedback itself gives NLR gas an additional boost. To probe this, provided we know the SMBH mass, we can for tight bins of masses plot the BPT dispersion again. If no AGN feedback is present, it should be almost uniform in velocity. If there is feedback, the AGN branch should show increased velocities despite having similar BH masses.

3.3 Black Hole Mass

To find the SMBH mass, we can use well known relations mapping stellar dispersion σ_* to central engine properties, provided that their velocities are predominantly determined by its gravitation (Tremaine et al. 2002, Gebhardt et al. 2000, Wu 2009). Since we do not have access to star velocities in the galactic cores of our sample, we instead rely on $\sigma_{[OIII]}$ as a proxy. As shown in Figure 14 this method reproduces our expectations from the velocity trend, but brings with it some serious flaws as well. For one, making use of such an approximation prevents us from determining the previously mentioned AGN feedback effects. Since $\sigma_{[OIII]}$ determines the mass, we cannot bin $M_{\rm BH}$ and expect to see increased kinematics above the gravitational baseline when we start from the fundamental assumption that the BH gravity itself is the sole culprit for the dispersion in the first place. Other than this circularity, there is also the issue of our galactic sources not meeting the underlying assumption. For SF galaxies, [OIII] emission occurs primarily inside the thin disk and not the bulge, meaning that velocities and theresulting BH mass will tend to be underestimated. The opposite case is true for AGN objects. Since gas is likely to be strongly broadened by effects other than gravity, we are biased towards assign unrealistically large SMBH masses. To compensate these shortcomings, we require another robust and independent way to determine velocities governed by gravity near the galactic core. Despite lacking such a quantity, for the next subsection, we will proceed as if our data already had been corrected in such a way, so that once we can access it, we will be able to easily repeat and extend these analysis steps with the new values. Figure 15 shows the BPT dispersion in four BH mass bins as indicated in the graphic. The change between subplots matches what we expect for gravitational rotation but shows no feedback signature.

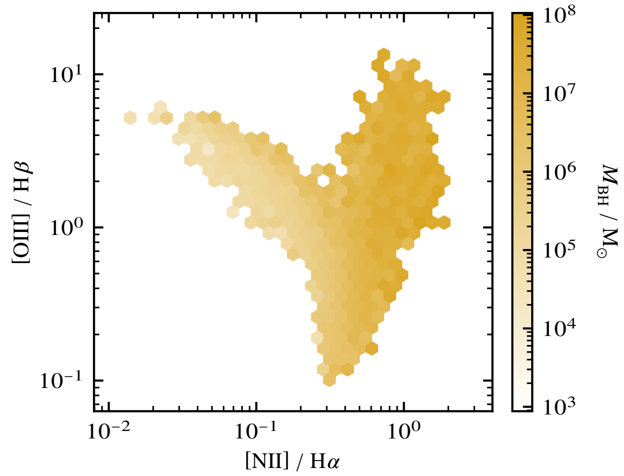

Figure 14: Binned BPT data showing black hole mass. Refer to Figures 22 and 23 for WHAN and WISE plots.

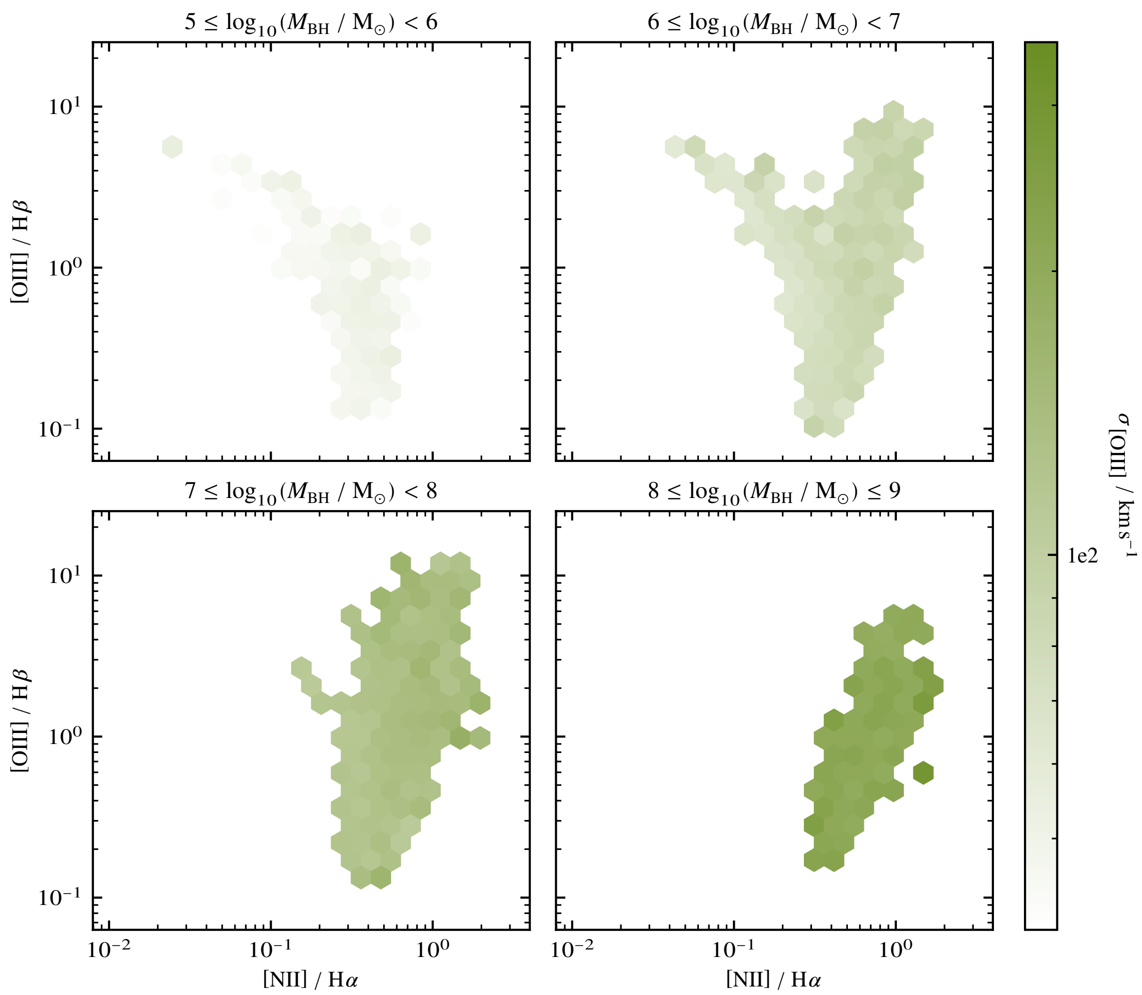

Figure 15: Binned BPT data of gas velocity for magnitude ranges of SMBH mass across subplots.

3.4 Data Analysis

Expanding on the preceding paragraph, binning in BH mass shows some gradient within bins, notably the 10^6 to 10^7 $M_\odot$ and 10^7 to 10^8 $M_\odot$ ranges. If we were selecting our mass steps more restrictively and had independent dispersion data, this would be an indicator for AGN feedback driven excess velocities that distringuish active cores from quiescent ones, but in our case it is simply an artifact of global scaling within broad mass bins. Another way we can isolate this effect is by plotting median dispersions based on our discrete BPT classes. Figure 16 does just that, and as expected we have an overall increase in velocities with BH mass but no significant stepwise increase towards faster motion in AGN vs SF types within individual bins.

Looking at figure 17 for a general population analysis, we can see that density and velocity show some positive correlation between each other. From the previous discussion on AGN driven compression and acceleration, this is physically sound. Contours for SF counts have a wider spread than COMP and AGN categories, and are centered at lower values for density as well as dispersion. As one might expect, COMP and AGN labels overlap more significantly, although AGN continue the trend of higher values on both axes with their distribution center. For BH masses, the same general observations hold as for electron density, which makes sense since the former derives directly from the latter. Especially for SF galaxies, we find very low masses from 10^3 to 10^4 $M_\odot$ which are telltale signs of systematic bias towards underestimation in objects dominated by emission lines from their stellar disks.

4 Conclusion

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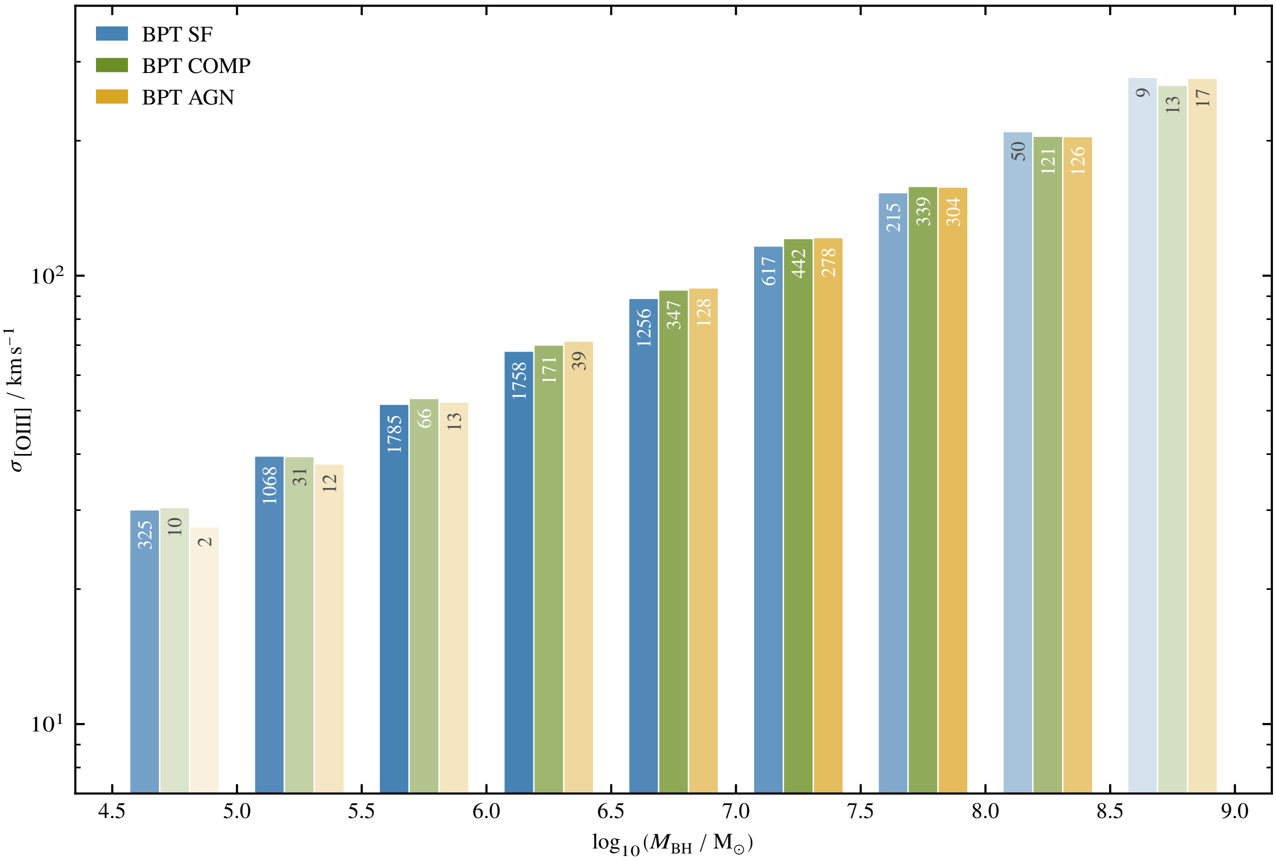

Figure 16: Bar chart of median gas dispersion by BPT class per BH mass bin. Numbers for members in bins of each class are given in the plot. Based on this, colors have increased saturation to represent higher statistical significance of the sample. The nearly perfect linear slope of the data is a consequence of how we generate BH masses from dispersion. In absence of statistical scatter, this would fit our starting function perfectly, meaning that we gain no real insight from this. Had we calculated BH mass in another independent way, this would however possibly reveal at which cutoff feedback plays an important role, if any, to determine the gas kinematics. Refer to Figures 24 and 25 for SDSS and WHAN classes.

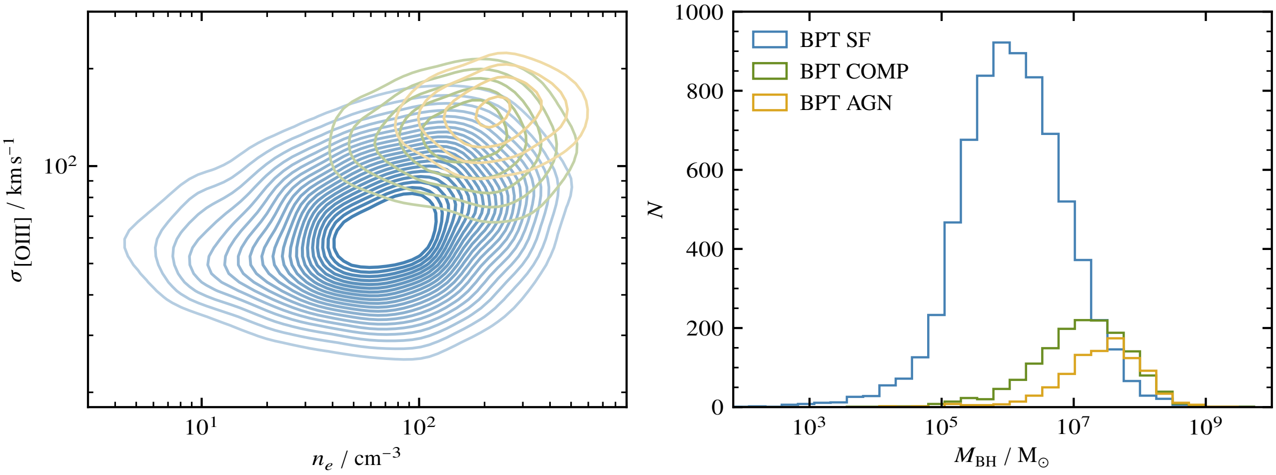

Figure 17

References

Baldwin, J. A. et al. *Publications of the Astronomical Society of the Pacific* (1981)
Cid Fernandes, R. et al. *Monthly Notices of the Royal Astronomical Society* (2011)
Gebhardt, K. et al. *The Astrophysical Journal* (2000)
Joh, K. et al. *Publications of the Astronomical Society of Japan* (2021)
Kauffmann, G. et al. *Monthly Notices of the Royal Astronomical Society* (2003)
Kewley, L. J. et al. *The Astrophysical Journal* (2001)
Kollmeier, J. A. et al. *The Astronomical Journal* (2026)
Mateos, S. et al. *Monthly Notices of the Royal Astronomical Society* (2012)
Sanders, R. L. et al. *The Astrophysical Journal* (2016)
Tremaine, S. et al. *The Astrophysical Journal* (2002)
Wright, E. L. et al. *The Astronomical Journal* (2010)
Wu, Q. *Monthly Notices of the Royal Astronomical Society* (2009)

Appendix

This section contains supplementary graphics as well as the SQL query in Figure 28 used to obtain our crossmatched dataset from this site. The full implementation with numerical values and equations via raw Python code is found in this repository.

A Gas Density

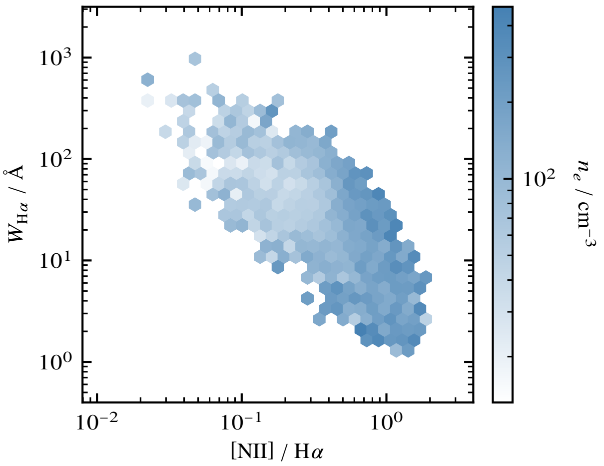

Figure 18: Binned WHAN data showing electron density.

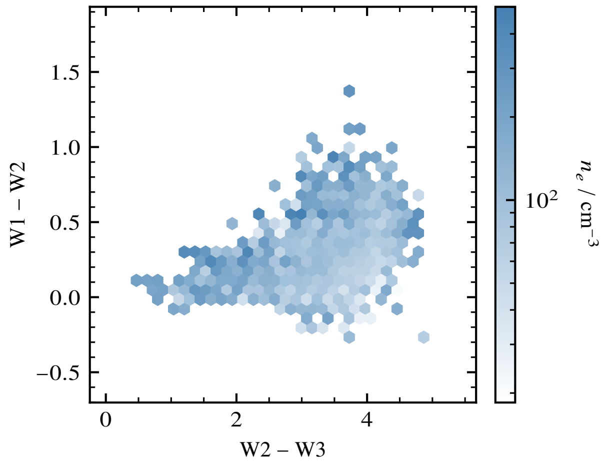

Figure 19: Binned WISE data showing electron density.

B Gas Velocity

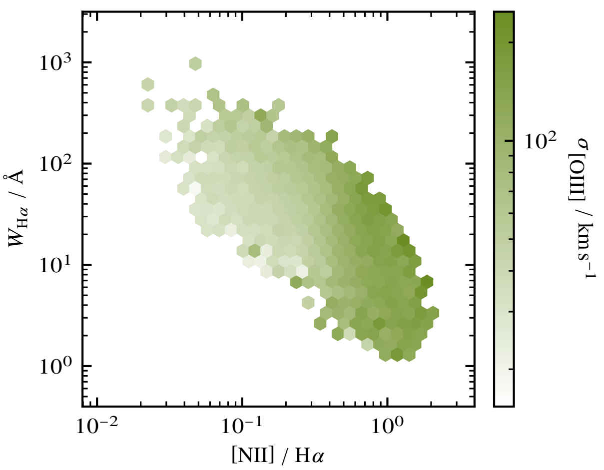

Figure 20: Binned WHAN data showing gas velocity.

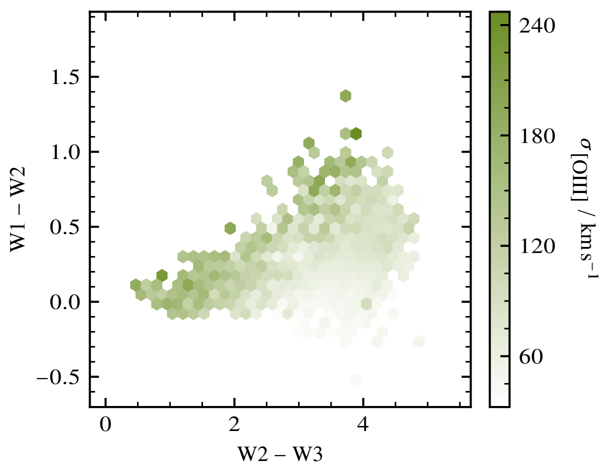

Figure 21: Binned WISE data showing gas velocity.

C Black Hole Mass

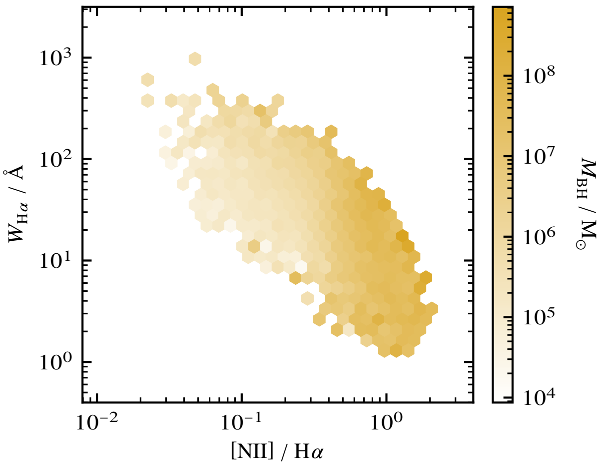

Figure 22: Binned WHAN data showing black hole mass.

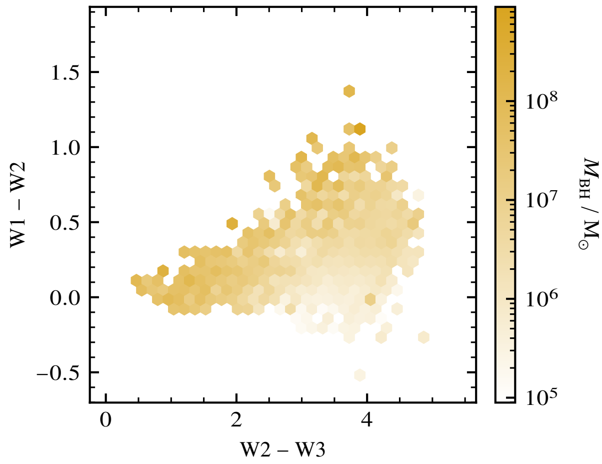

Figure 23: Binned WISE data showing black hole mass.

D Data Analysis

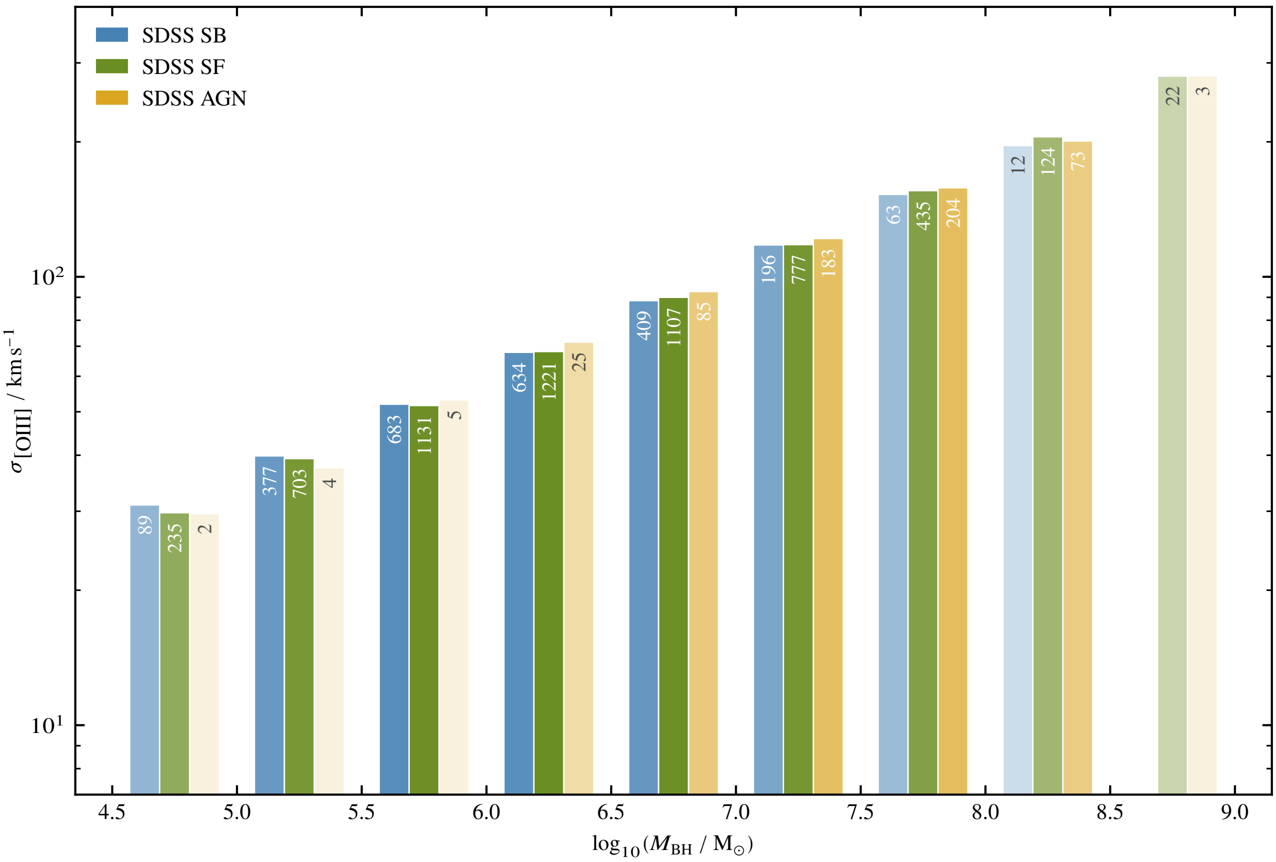

Figure 24: Bar chart of median gas dispersion by SDSS class per BH mass bin. Numbers for members in bins of each class are given in the plot. Based on this, colors have increased saturation to represent higher statistical significance of the sample. The nearly perfect linear slope of the data is a consequence of how we generate BH masses from dispersion. In absence of statistical scatter, this would fit our starting function perfectly, meaning that we gain no real insight from this. Had we calculated BH mass in another independent way, this would however possibly reveal at which cutoff feedback plays an important role, if any, to determine the gas kinematics.

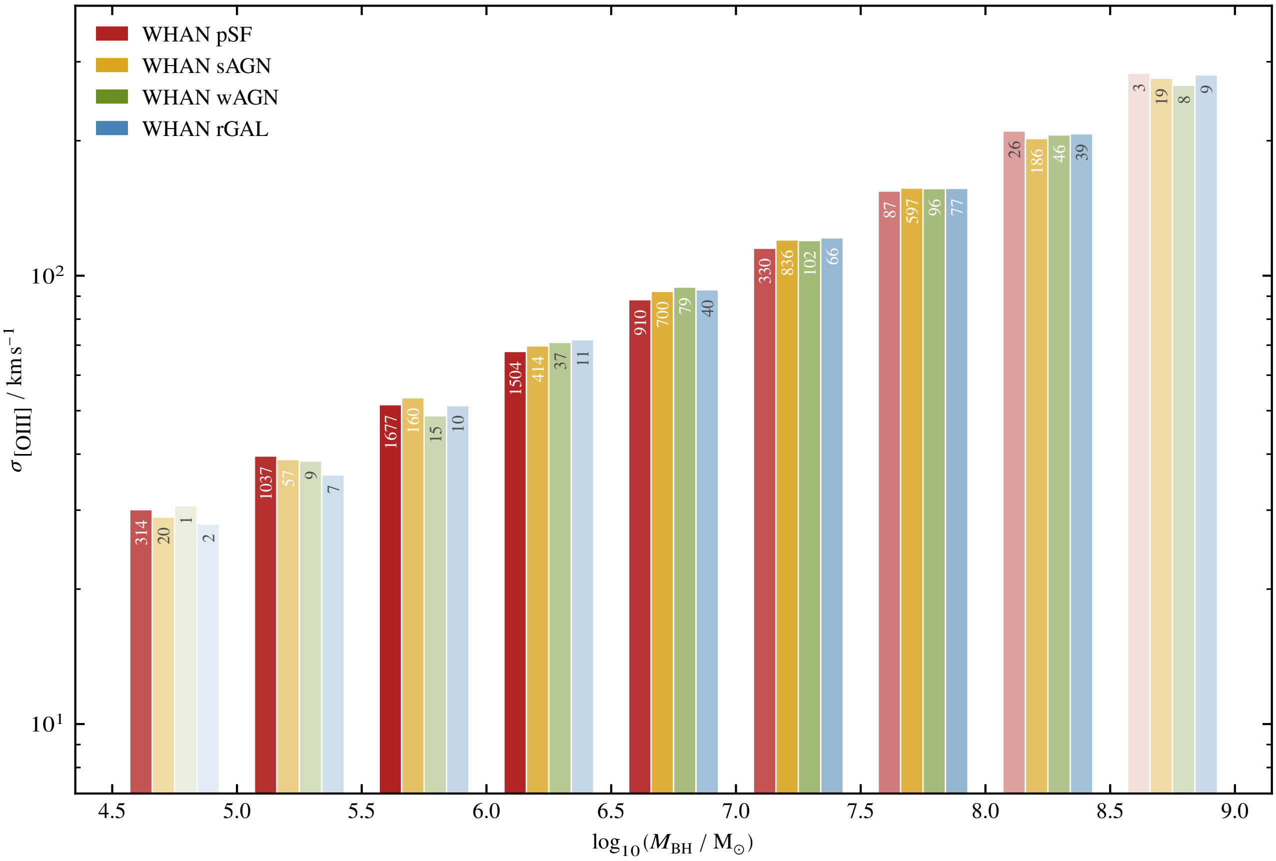

Figure 25: Bar chart of median gas dispersion by WHAN class per BH mass bin. Numbers for members in bins of each class are given in the plot. Based on this, colors have increased saturation to represent higher statistical significance of the sample. The nearly perfect linear slope of the data is a consequence of how we generate BH masses from dispersion. In absence of statistical scatter, this would fit our starting function perfectly, meaning that we gain no real insight from this. Had we calculated BH mass in another independent way, this would however possibly reveal at which cutoff feedback plays an important role, if any, to determine the gas kinematics.

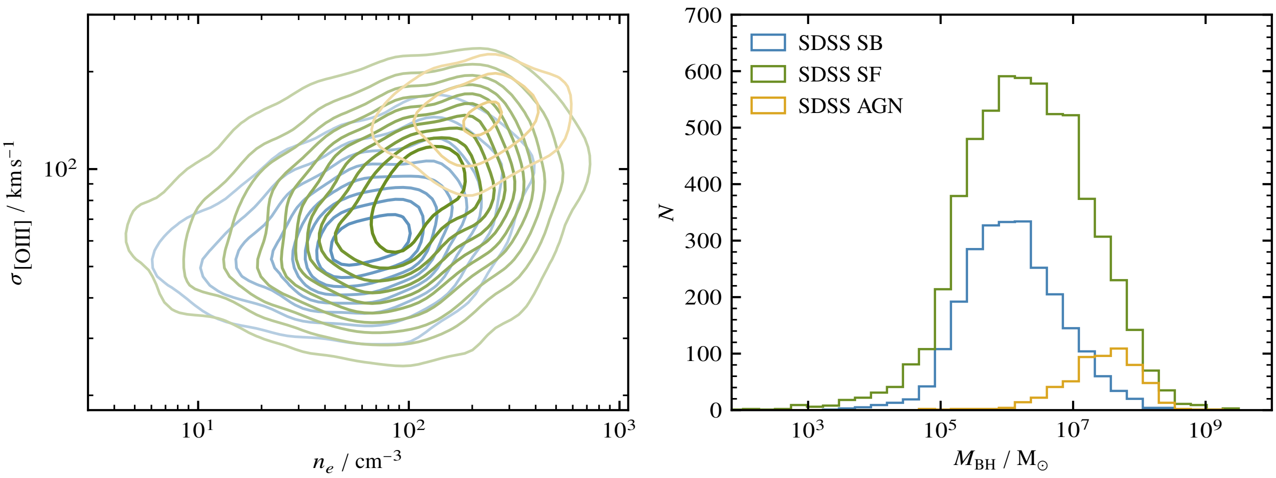

Figure 26

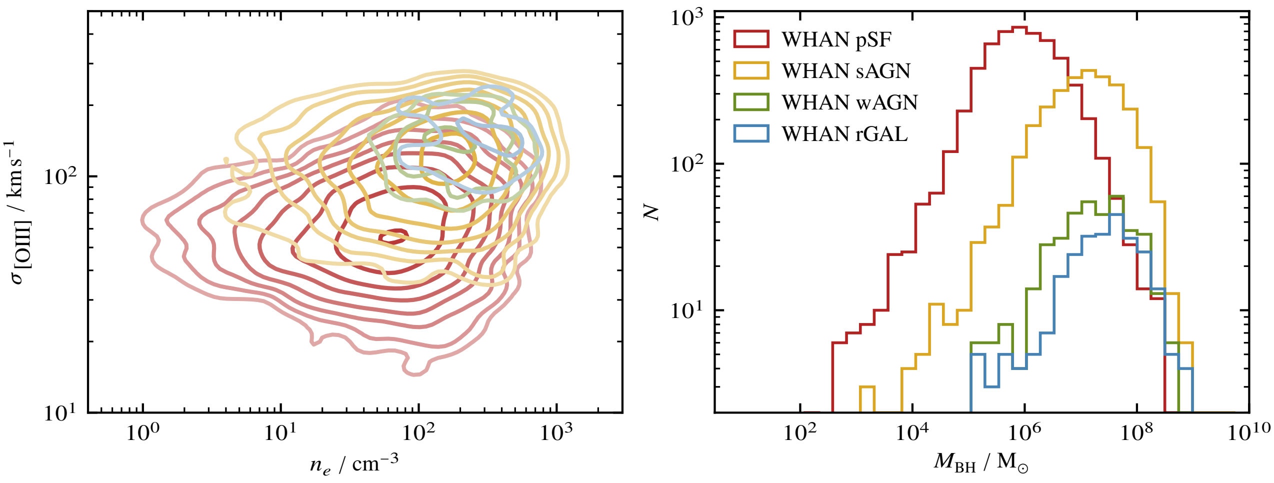

Figure 27

E Database Query

```sql
SELECT TOP 10000
    g.sii_6717_flux, g.sii_6731_flux, g.nii_6584_flux, g.oi_6300_flux, g.oiii_5007_flux,
    g.h_alpha_flux, g.h_beta_flux,
    g.h_alpha_eqw, g.nii_6584_eqw, g.oiii_sigma,
    w.w1mpro, w.w2mpro, w.w3mpro,
    s.z, s.class, s.subclass, s.bestobjid
FROM SpecObj AS s
    JOIN galSpecLine AS g ON s.specobjid = g.specobjid
    JOIN PhotoTag AS p ON s.bestobjid = p.objid
    OUTER APPLY (   SELECT TOP 1 x.wise_cntr
                    FROM wise_xmatch AS x
                    WHERE p.objid = x.sdss_objid AND x.match_dist < 3
                    ORDER BY x.match_dist ASC
                ) AS y
    LEFT JOIN wise_allsky AS w ON y.wise_cntr = w.cntr
WHERE s.class = 'GALAXY'
    AND s.subclass != 'BROADLINE'
    AND s.subclass != 'AGN BROADLINE'
    AND s.subclass != 'STARFORMING BROADLINE'
    AND s.subclass != 'STARBURST BROADLINE'
    AND s.snmedian_r > 5.0
    AND s.z < 0.35
    AND g.sii_6717_eqw < 0.0
    AND g.sii_6731_eqw < 0.0
    AND g.nii_6584_eqw < 0.0
    AND g.oi_6300_eqw < 0.0
    AND g.oiii_5007_eqw < 0.0
    AND g.h_alpha_eqw < 0.0
    AND g.h_beta_eqw < 0.0
    AND (2.355 * g.sigma_forbidden) < 500
    AND (2.355 * g.sigma_balmer) < 500
    AND g.sii_6717_flux > 3 * g.sii_6717_flux_err
    AND g.sii_6731_flux > 3 * g.sii_6731_flux_err
    AND g.nii_6584_flux > 3 * g.nii_6584_flux_err
    AND g.oi_6300_flux > 3 * g.oi_6300_flux_err
    AND g.oiii_5007_flux > 3 * g.oiii_5007_flux_err
    AND g.h_alpha_flux > 3 * g.h_alpha_flux_err
    AND g.h_beta_flux > 3 * g.h_beta_flux_err
ORDER BY s.specobjid ASC
```

Figure 28